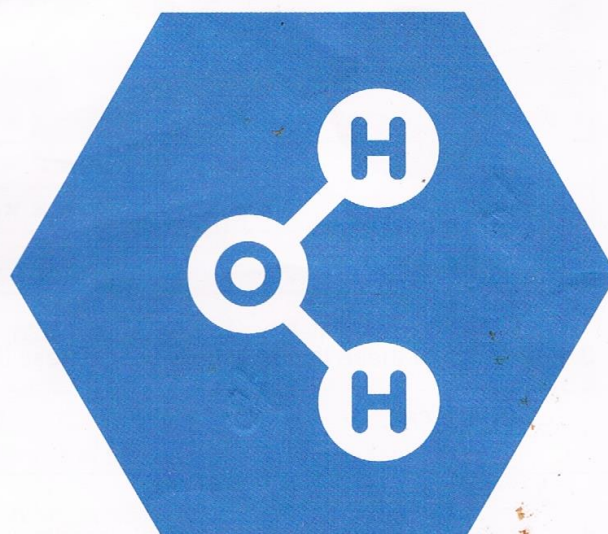




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

INTRODUCTORY PAPER



2014 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1-30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

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1. Which one of these animals has wings?

- (A) a bat (B) a shark (C) a mouse (D) a snake

2. Which of these will bounce highest when dropped onto a floor?



(A) a basketball



(B) a book



(C) a shoe

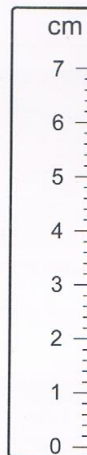


(D) a sock

3. Which of these things usually weighs less than a bicycle?

- (A) a car (B) a cat (C) a house (D) a cow

4. Milan measures the height of this candle with a ruler.



What is the height of the candle?

- (A) 7 cm (B) 6 cm (C) 4 cm (D) 1 cm

5. When is it easiest to see lots of stars if there are no clouds?

- (A) sunrise (B) midday (C) sunset (D) midnight

6. Simon helped clean up a park on three days.

The table shows how much of each kind of rubbish he found.

x = 1 piece of rubbish

	Bottles	Cans	Plastic bags	Paper
Day 1	x x x x	x x x	x x	x
Day 2	x x	x x x x x	x x x	x
Day 3	x x x	x x x	x x x	x x x x x x

Which type of rubbish did he find the most of?

- (A) Bottles (B) Cans (C) Plastic bags (D) Paper

7. Diana wants to know the kind of pet that people in her school like most.

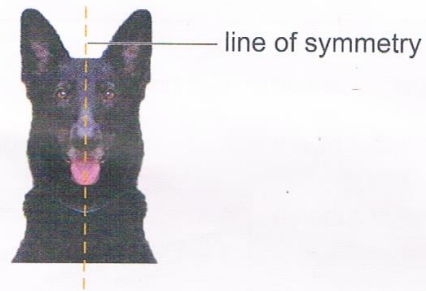
She does not have time to ask everybody in her school.

Which of these would be the best way to find out?

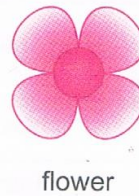
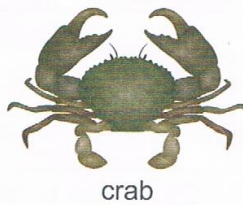
- (A) Ask her friends what their favourite kind of pet is.
(B) Ask only girls what their favourite kind of pet is.
(C) Ask only younger students what their favourite kind of pet is.
(D) Ask some boys and girls from each year group what their favourite kind of pet is.

8. Many living things are symmetrical.

For example, this picture of a dog has one line of symmetry.



The pictures below show a crab and a flower.



Which option has the correct number of lines of symmetry for each picture?

	crab	flower
(A)	1	2
(B)	2	2
(C)	1	4
(D)	2	4

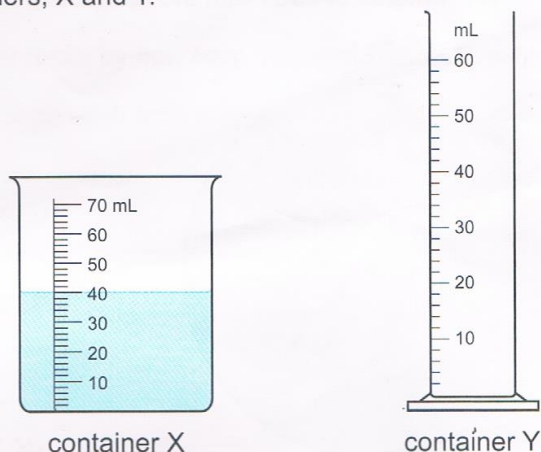
9. The table lists the features of some different shapes.

Shape	Colour	Size
square	orange	medium
circle	blue	large
rectangle	red	medium
square	orange	small
circle	yellow	small

What colour is the small square shape?

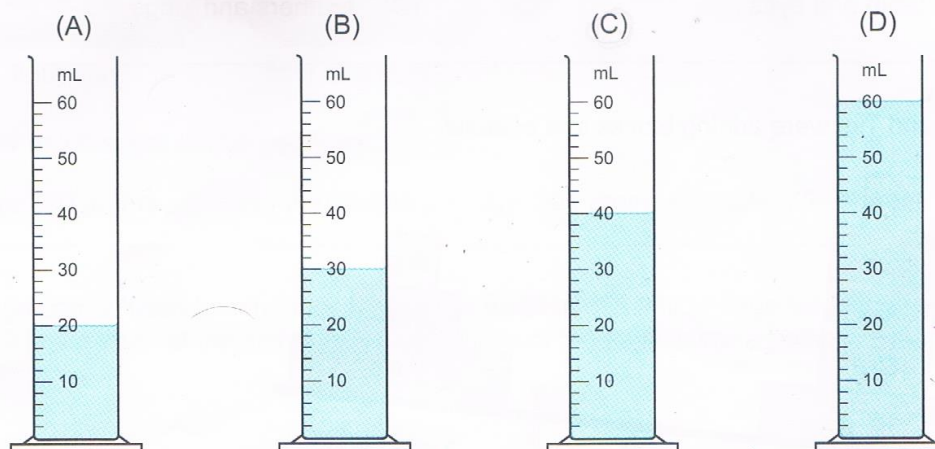
- (A) orange (B) blue (C) red (D) yellow

10. Joe had two containers, X and Y.



All the water from container X was poured into container Y.

Which diagram shows the water level in container Y?



11. Dave looked at four machines in his kitchen.

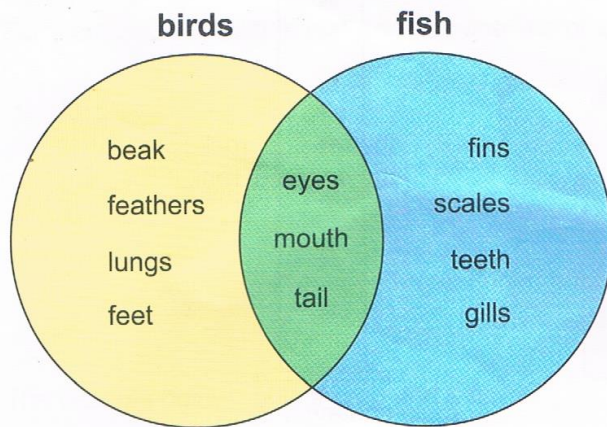
Each machine changed electrical energy into other types of energy.

		Fridge	Grill	Oven	Blender
Type of energy	sound	✓			✓
	light	✓	✓	✓	
	heat	✓	✓	✓	✓
	movement				✓

Which type of energy do all the machines change electrical energy into?

- (A) sound (B) light (C) heat (D) movement

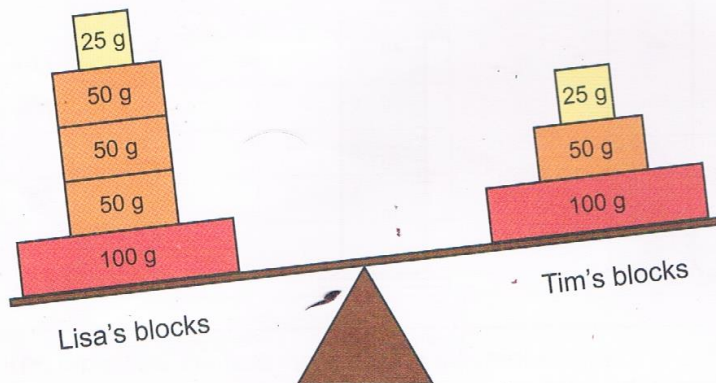
12. The diagram compares the features of birds with the features of fish.



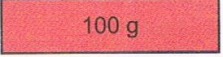
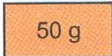
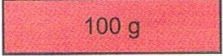
What are two features that birds and fish have in common?

- (A) mouth and tail
(B) fins and teeth
(C) teeth and eyes
(D) feathers and lungs

13. Lisa and Tim were adding blocks to a seesaw.



Which block(s) should Tim add to his side to balance the seesaw?

- (A)  + 
- (B)  + 
- (C)  + 
- (D) 

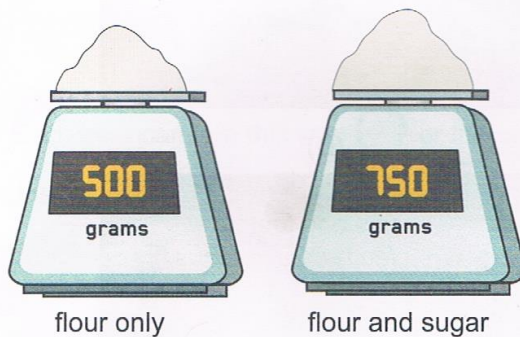
14. Alice collects rocks.

She wants to sort her rocks by how hard they are. She tests some rocks.

What should she change each time she tests a rock?

- (A) the length of rock (B) the mass of rock
(C) the type of rock (D) the shape of rock

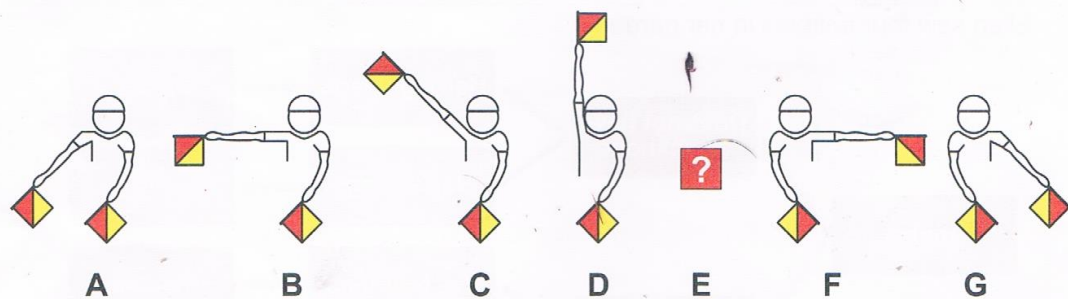
15. Brian put some flour on a scale. He then added some sugar to the flour.



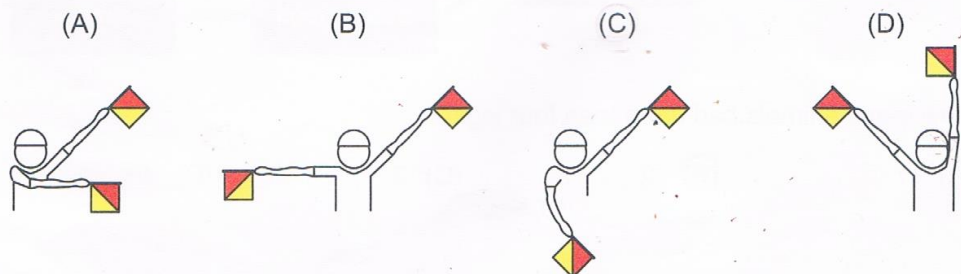
How much sugar did he add?

- (A) 100 grams (B) 250 grams (C) 500 grams (D) 750 grams

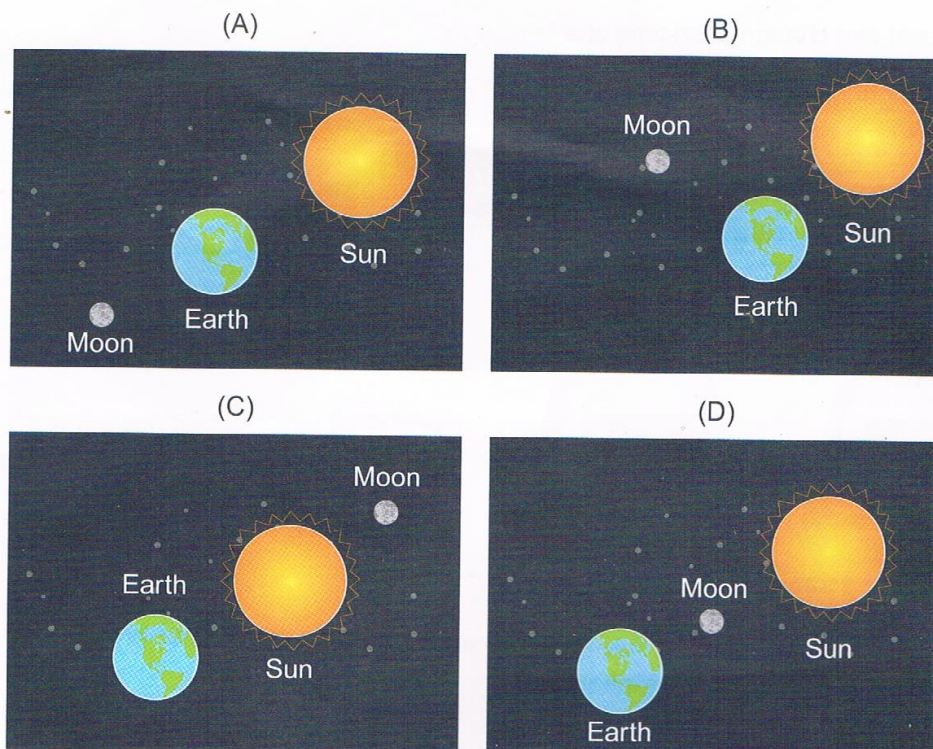
16. Flags can be used to send messages. The position of a pair of flags held by a person matches a letter of the alphabet. The positions of the flags follow a pattern.



Which of the following represents the letter E?

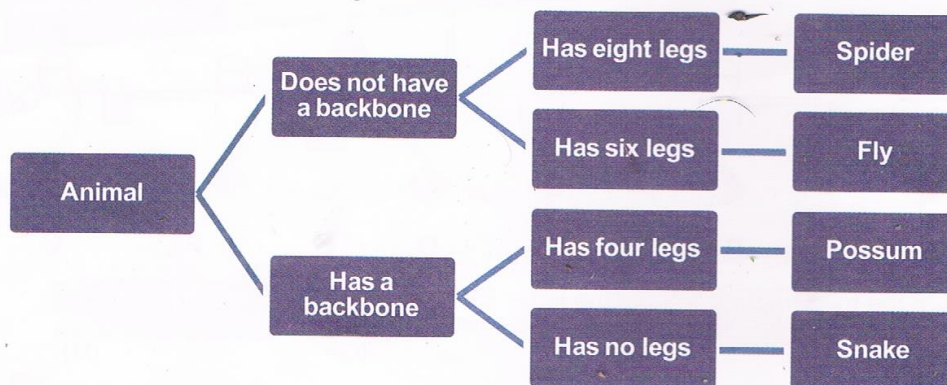


17. A solar eclipse occurs when the Moon blocks some of the sunlight from reaching Earth.
Which diagram shows a solar eclipse?



(NOT TO SCALE)

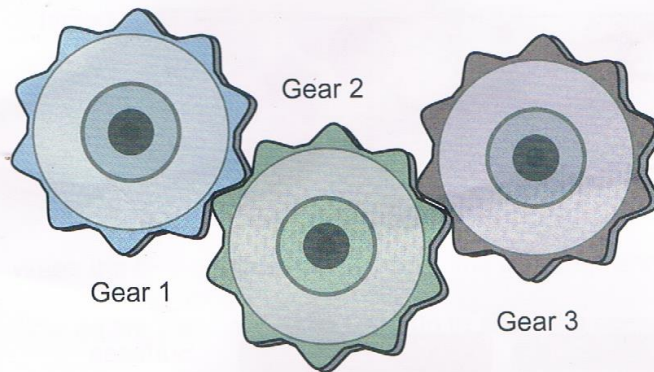
18. Ellen saw four animals in her garden.



How many animals had more than four legs?

- (A) 1 (B) 2 (C) 3 (D) 4

19. The picture shows three gears.



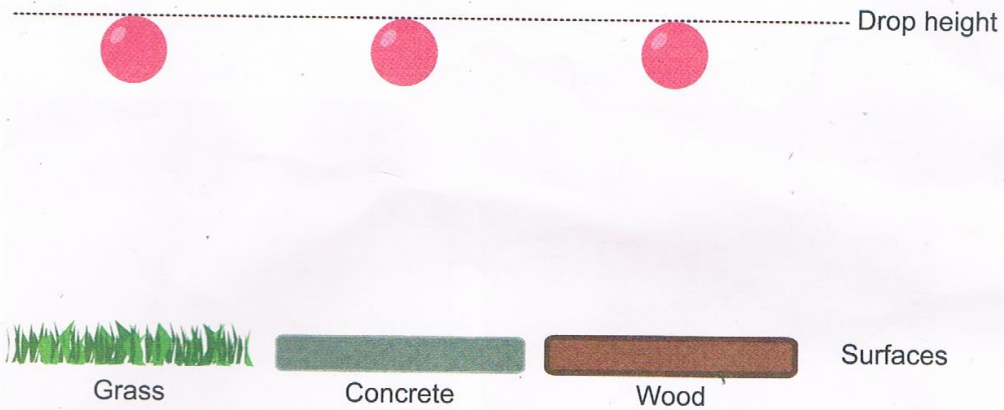
Each gear can turn this way  or this way .

Joe turns Gear 1 which makes the other gears turn.

Which of these shows the directions the gears could be turning?

	Gear 1	Gear 2	Gear 3
(A)			
(B)			
(C)			
(D)			

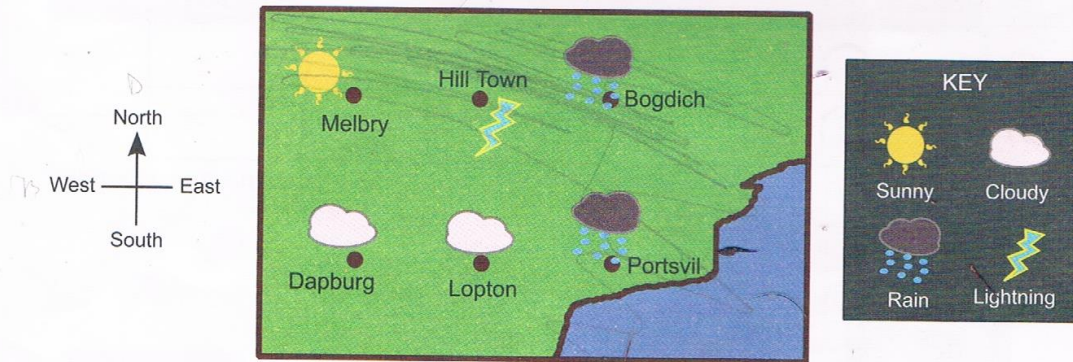
20. Sam was testing the bounce of a rubber ball. The picture shows three of his tests.



What can Sam find out from these three tests?

- (A) whether the type of ball affects the bounce
- (B) whether the type of surface affects the bounce
- (C) whether the drop height affects the bounce
- (D) whether the ball shape affects the bounce

21. This is a weather map for six towns.



Danny lives in a town that is west of Bogdich and directly north of Dapburg.

What weather does the map show for Danny's town?

- (A) Sunny
- (B) Cloudy
- (C) Rain
- (D) Lightning

22. The diagrams show four tiles, each with part of a plant drawn on it.



tile 1



tile 2



tile 3

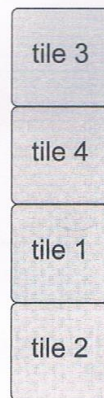


tile 4

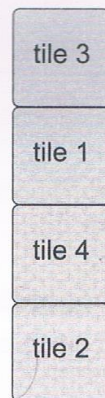
When the tiles are lined up correctly, the drawings show the complete plant.

How do the tiles have to be lined up to show the complete plant?

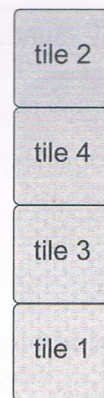
(A)



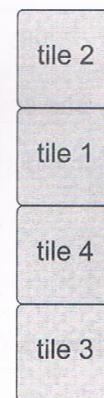
(B)



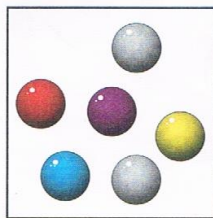
(C)



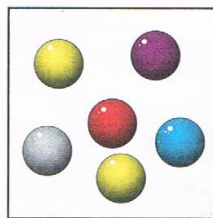
(D)



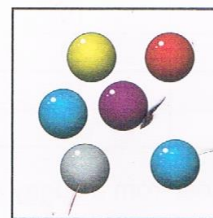
23. Lisa makes five groups of coloured marbles that follow a pattern.



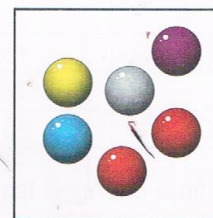
Group 1



Group 2



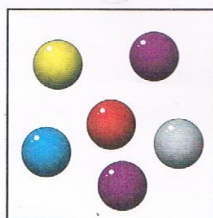
Group 3



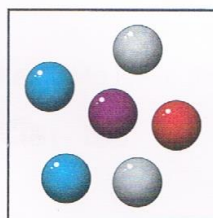
Group 4

Which of the following is the fifth group of marbles?

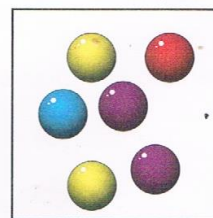
(A)



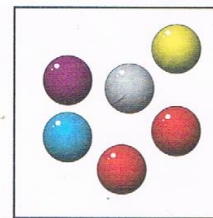
(B)



(C)

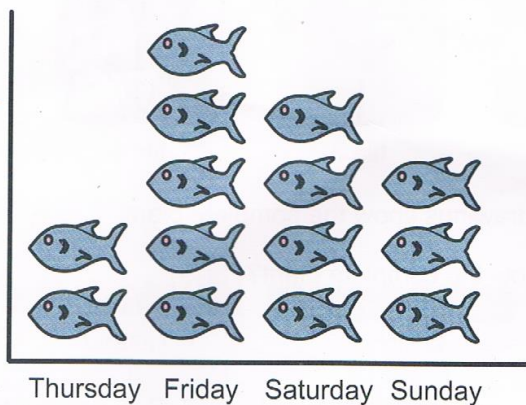


(D)



24. Sarah counted the number of fish she saw in a river over four days.

She drew this graph to show the number she saw.



On Saturday she saw 8 fish.

How many fish did she see altogether?

25. Four children wrote down the time it took them to get to school on a sunny day and on a rainy day.

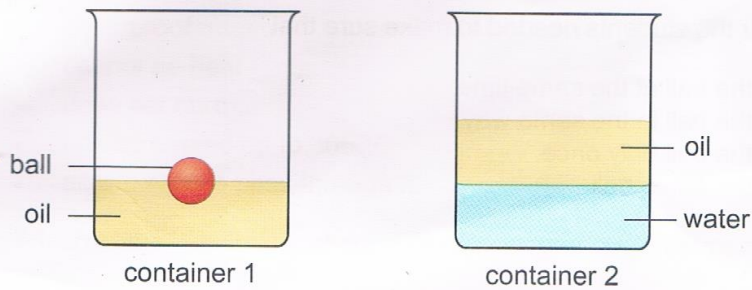
The table shows their times in minutes.

	Sunny day	Rainy day
Amy	10	15
Ben	15	20
Clara	5	15
David	20	20

Whose time changed the most from a sunny day to a rainy day?

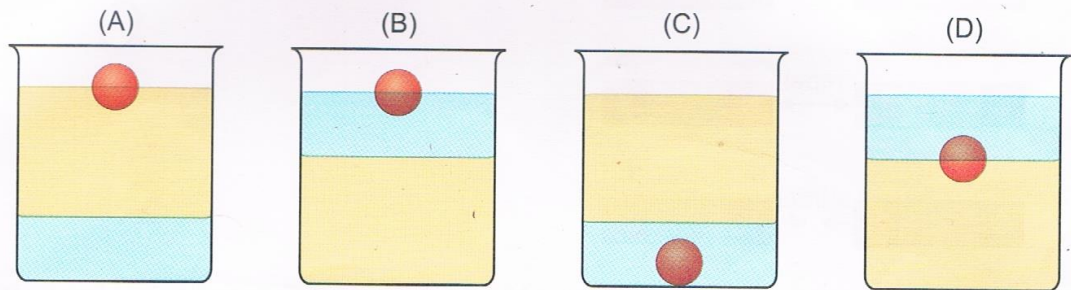
- (A) Amy (B) Ben (C) Clara (D) David

26. Container 1 had oil and a small rubber ball in it. Container 2 had water and oil in it.



Jane poured the contents of container 2 into container 1.

Which diagram shows what container 1 looks like now?



27. Matt was walking along the sand when he saw these footprints.



Which animal was the last to leave these footprints on the sand?

- (A) cat (B) bird (C) human (D) horse

28. Some students played a game to see who could throw a ball the furthest.

For their game to be fair the students needed to make sure that

- (A) they each threw the ball at the same time.
- (B) they each threw the ball in the same way.
- (C) they each threw the ball only once.
- (D) they each threw the same ball.

29. Magnets have a north pole and a south pole. When magnets are placed near one another, they can either repel or attract one another as shown in the diagram.



Four students were given the challenge of joining four magnets together in a line so that they would remain joined.

Which line of magnets will join together?

- (A)
- (B)
- (C)
- (D)

30. Scientists made a new material with these features:

- waterproof
- cannot be bent
- does not burn
- can be heated up to 300 °C
- allows heat to pass through it.

Which of these is the most suitable product that could be made using this new material?

- (A) oven gloves
- (B) garden hose
- (C) drinking glass
- (D) saucepan

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Sources

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PRACTICE QUESTION

Which of these animals lays eggs?

- (A) dog
- (B) rabbit
- (C) horse
- (D) chicken

Turn your answer sheet over and fill in the correct oval.

The following year levels should sit THIS Paper:

Australia

Year 2



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ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Interpret a picture graph about fish	Interpreting	24	D
	Determine the number of lines of symmetry for given objects	Observing/Measuring	8	C
	Determine what needs to be done to keep a test fair	Investigating	28	D
	Recognise similarity between groups	Reasoning/Problem solving	23	A
	Select a consistent set of turns for three gears	Reasoning/Problem solving	19	C
	Recognise the relative density of a solid and two liquids	Reasoning/Problem solving	26	A
	Determine the best attribute to change when testing some rocks	Investigating	14	C
	Sequence the vertical order of tiles to complete a picture	Observing/Measuring	22	B
	Read a weather map	Interpreting	21	A
	Sequence the relative time of different footprints	Reasoning/Problem solving	27	A
	Select the correct masses to balance a see-saw	Observing/Measuring	13	D
	Predict the potential use of a product from its properties	Reasoning/Problem solving	30	D
	Predict a letter using the flags in a semaphore system	Predicting/Concluding	16	C
	Identify the greatest change in a set of measurements due to a change in conditions	Interpreting	25	C
	Identify the attributes of magnets	Reasoning/Problem solving	29	B
Easy Questions	Identify what can be discovered from a set of tests	Investigating	20	B
	Recognise that the volume of a liquid remains constant	Predicting/Concluding	10	C
	Use a key to find a quantity	Interpreting	18	B
	Calculate the size of a change in mass	Predicting/Concluding	15	B
	Determine the conditions for a solar eclipse	Interpreting	17	D
	Identify the largest category from a table with multiple rows and columns	Interpreting	6	B
	Identify common features of animal groups from a Venn diagram	Predicting/Concluding	12	A
	Determine which classification is common to four sets of data	Interpreting	11	C
	Identify the best way of collecting data from a group of people	Investigating	7	D
	Extract information from a table	Predicting/Concluding	9	A
	Estimate which object weighs less than a bicycle	Observing/Measuring	3	B
	Identify a time of day when stars are most visible	Observing/Measuring	5	D
	Measure the height of a candle	Observing/Measuring	4	C
	Identify which of a set of objects will bounce highest	Predicting/Concluding	2	A
	Recognise common features of animals such as wings	Interpreting	1	A